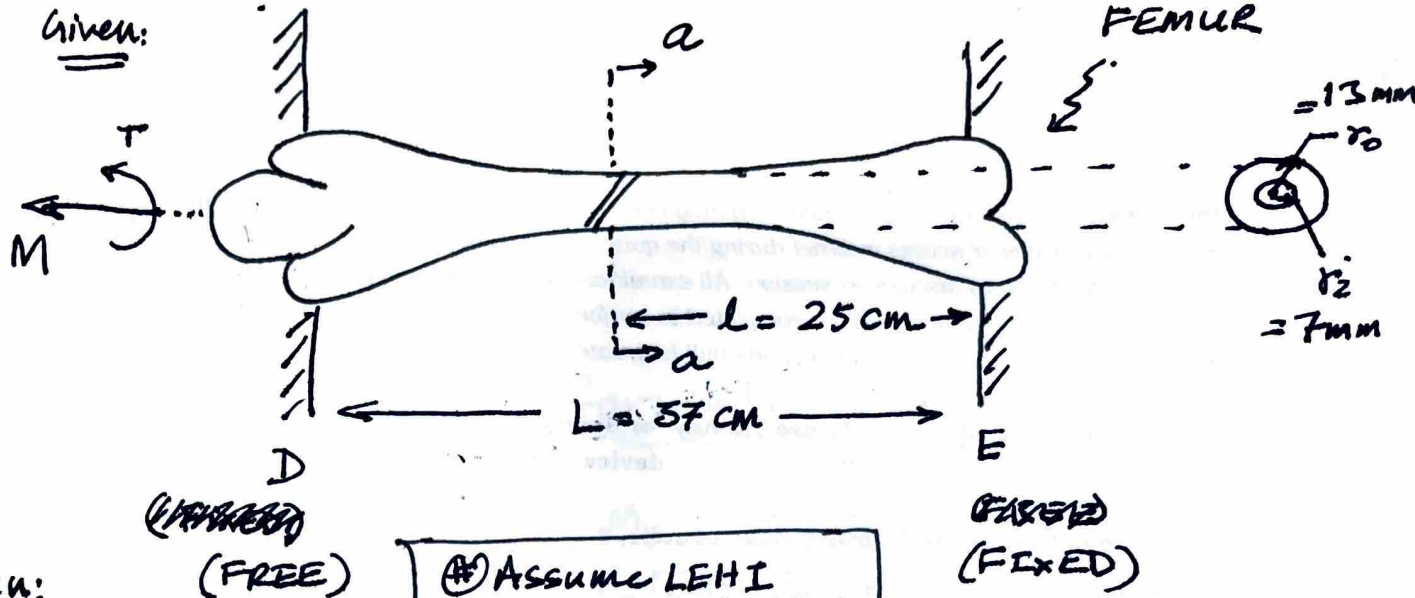


# Problem - 1:

March 24, 2026 KVB

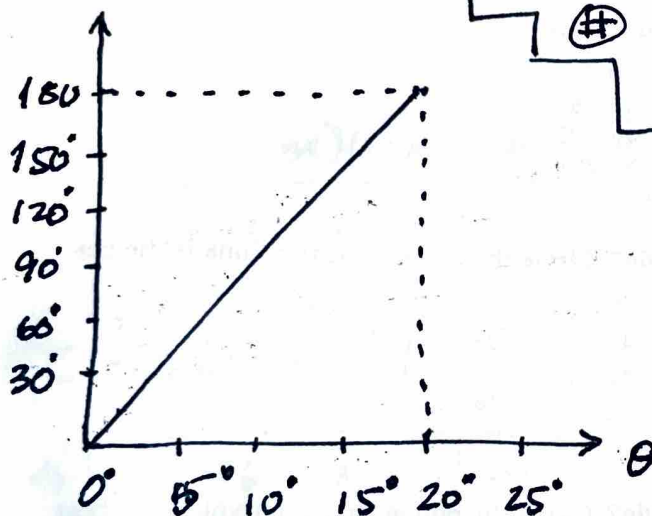
①

① Femur is subjected to a torsional Loading until fracture



Given:

$M \text{ (N-m)}$



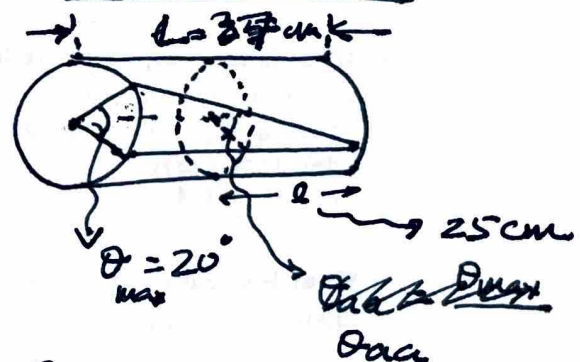
{ Torque applied vs Angle of twist }

① Assume LEH I behavior

① Maximum angle of twist before fracture:  $\theta_{\max} = 20^\circ$   
( $\theta$  at ~~aa~~ - section, max)

① Shear strain ( $\gamma$ ) at surface of the bone:

$$\gamma = \frac{r_o}{L} \theta_{aa}$$



Calculate ① Maximum shear strain

② Maximum shear stress & ③ Shear Modulus of elasticity of Femur

$13.5^\circ$  in radians

$$(13.5^\circ) \left( \frac{\pi}{180} \right)$$

$\approx 0.236$  radians

Angular deformation

(max. twist per unit axial length)

$$\frac{\theta_{\max}}{L} = \frac{20^\circ}{37 \text{ cm}} = 0.54^\circ/\text{cm}$$

$$\left( \text{angular deformation at section a-a} \right) = \left( \frac{0.54^\circ}{\text{cm}} \right) (25 \text{ cm}) = 13.5^\circ$$

Shear strain at section a-a

$$\gamma_{a-a} = \frac{r_o}{l} \theta_{aa} \Rightarrow \gamma_{aa} = \left( \frac{0.3 \times 10^{-3} \text{ m}}{25 \times 10^{-3} \text{ m}} \right) (0.236 \text{ radians})$$

$$\Rightarrow \gamma_{aa} = 0.0123 \text{ rad}$$

Torsional stress formula (At section a-a)  
(At fracture)

$$\tau = \frac{M r_o}{J}$$

$M_{\text{fracture}} = 180 \text{ N}\cdot\text{m}$   
 $r_o = 13 \times 10^{-3} \text{ m}$   
polar moment of inertia

$$\tau = \frac{(180 \text{ N}\cdot\text{m})(13 \times 10^{-3} \text{ m})}{41.1 \times 10^{-9} \text{ m}^4} \quad J = \frac{\pi}{2} (r_o^4 - r_i^4) = \frac{\pi}{2} ([13 \times 10^{-3}]^4 \text{ m}^4 - [7 \times 10^{-3}]^4 \text{ m}^4)$$

$$\tau = 56.9 \times 10^6 \text{ Pa} = 41.1 \times 10^{-9} \text{ m}^4$$

$$\Rightarrow \tau = 56.9 \text{ MPa} \rightarrow \tau = \tau_{\text{fracture}}$$

Shear Modulus (Assume LEIT)

$$\tau = G \gamma \Rightarrow G = \frac{\tau}{\gamma} = \frac{\tau_{\text{fracture}}}{\gamma_{aa}}$$

$$\Rightarrow G = \frac{56.9 \times 10^6 \text{ Pa}}{0.0123 \text{ rad}} = 4.6 \times 10^9 \text{ Pa}$$

$$\Rightarrow G = 4.6 \text{ GPa}$$

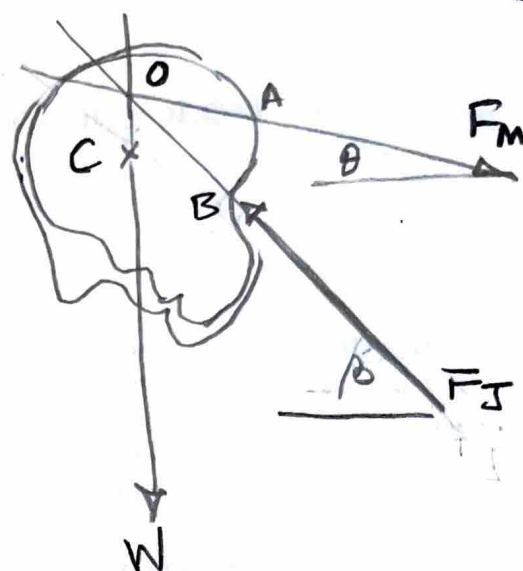
# HEAD & NECK

## PROBLEM 2

# HEAD & NECK ARE IN EQUILIBRIUM AS SHOWN

# WE HAVE A CONCURRENT FORCE SYSTEM AS SHOWN

# C  $\rightarrow$  center of gravity of the head



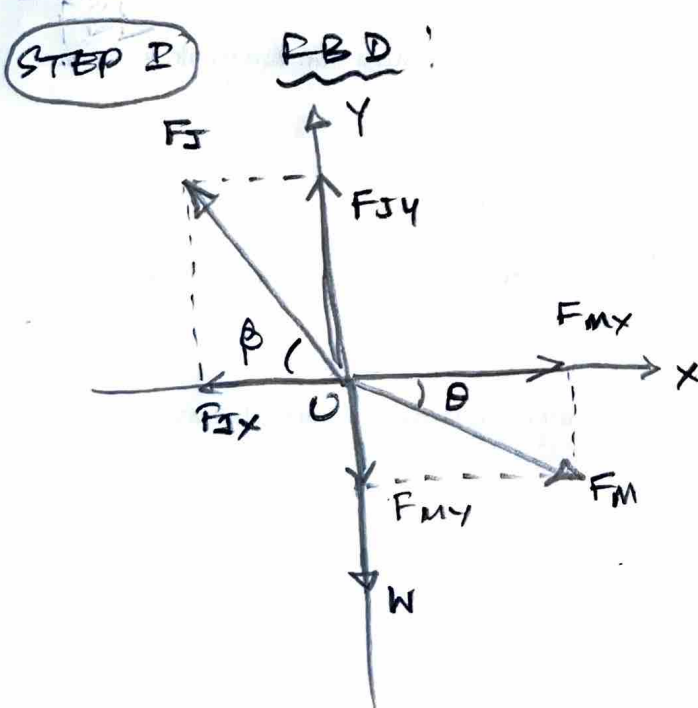
#  $F_m \rightarrow$  Force exerted by neck extensor muscles on the head at A.  $F_m = 57\text{ N}$   $\theta = 36^\circ$

#  $F_j \rightarrow$  Reaction force due to atlantooccipital joint at the center of rotation B  $\beta = 63^\circ$

1 Determine the magnitude of gravitational force acting on the head. 2 Magnitude of Reaction force

SOLUTION:

STEP II Resolve all forces.



$$\begin{aligned} F_{mx} &= F_m \cos \theta \\ F_{my} &= F_m \sin \theta \\ F_{jx} &= F_j \cos \beta \\ F_{jy} &= F_j \sin \beta \end{aligned}$$

$\rightarrow$  ①



(4)

STEP III

Apply Equilibrium conditions

$$\sum F_x = 0$$

$$-F_{Jx} + F_{mx} = 0 \Rightarrow F_{Jx} = F_{mx} \rightarrow (2)$$

$$\sum F_y = 0$$

$$-W - F_{my} + F_{Jy} = 0 \Rightarrow W = F_{Jy} - F_{my} \rightarrow (3)$$

From Eqs. (1) &amp; (2) we have

$$F_J \cos \beta = F_m \cos \theta \rightarrow (4)$$

From Eqs (1) &amp; (3) we have

$$W = F_J \sin \beta - F_m \sin \theta \rightarrow (5)$$

From (4) &amp; (5)

$$W = \frac{F_m \cos \theta}{\cos \beta} \sin \beta - F_m \sin \theta$$

$$W = F_m \cos \theta \tan \beta - F_m \sin \theta$$

$$\Rightarrow W = 57 \cos 36^\circ \tan 63^\circ - 57 \sin 36^\circ$$

$$57 [(0.809)(1.962) - (0.5877)]$$

$$\Rightarrow W = 57 \text{ N} \leftarrow \text{Ans (1)}$$

From (4)

$$F_J = \frac{F_m \cos \theta}{\cos \beta}$$

$$\Rightarrow F_J = \frac{57 \cos 36^\circ}{\cos 63^\circ}$$

$$\Rightarrow F_J = 101.58 \text{ N}$$

Ans (2)

OR

From (2)

$$F_{Jx} = F_{mx}$$

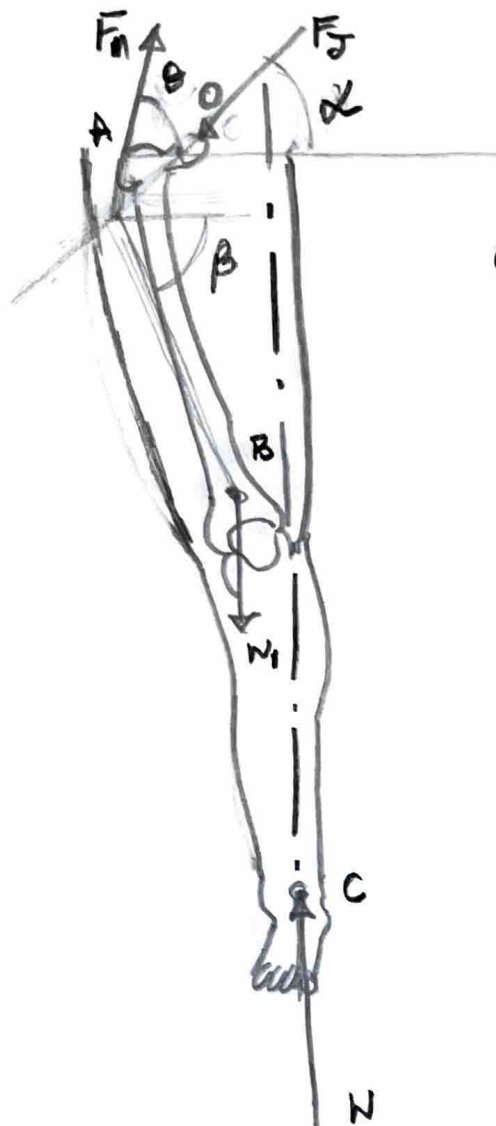
$$= F_m \cos \theta$$

$$= 57 \cos 36^\circ$$

$$= 46.113 \text{ N}$$

$$F_J = \frac{F_{Jx}}{\cos \beta}$$

$$\Rightarrow F_J = 101.58 \text{ N}$$



(#) ENTIRE WEIGHT OF A BODY IS PLACED ON ONE LEG AS SHOWN

(#) O  $\rightarrow$  center of rotation of the hip joint

(#)  $F_M \rightarrow$  Force exerted by the hip abductor muscles attached at point A

(#)  $W_1 \rightarrow$  Gravitational force acting at the center of gravity B

(#)  $F_J \rightarrow$  Magnitude of reaction force at the hip joint location O

(#)  $W \rightarrow$  Total weight of the body applied as a normal force from the ground.

Determine

① Total weight ( $W$ ) of the body

② Magnitude of  $F_J$  acting at the hip joint

Given: Distances:

$$\overline{AO} = a = 8.6 \text{ cm}$$

$$\overline{AB} = b = 34.3 \text{ cm}$$

$$\overline{AC} = c = 89.4 \text{ cm}$$

$$\rightarrow W_1 = 125 \text{ N}$$

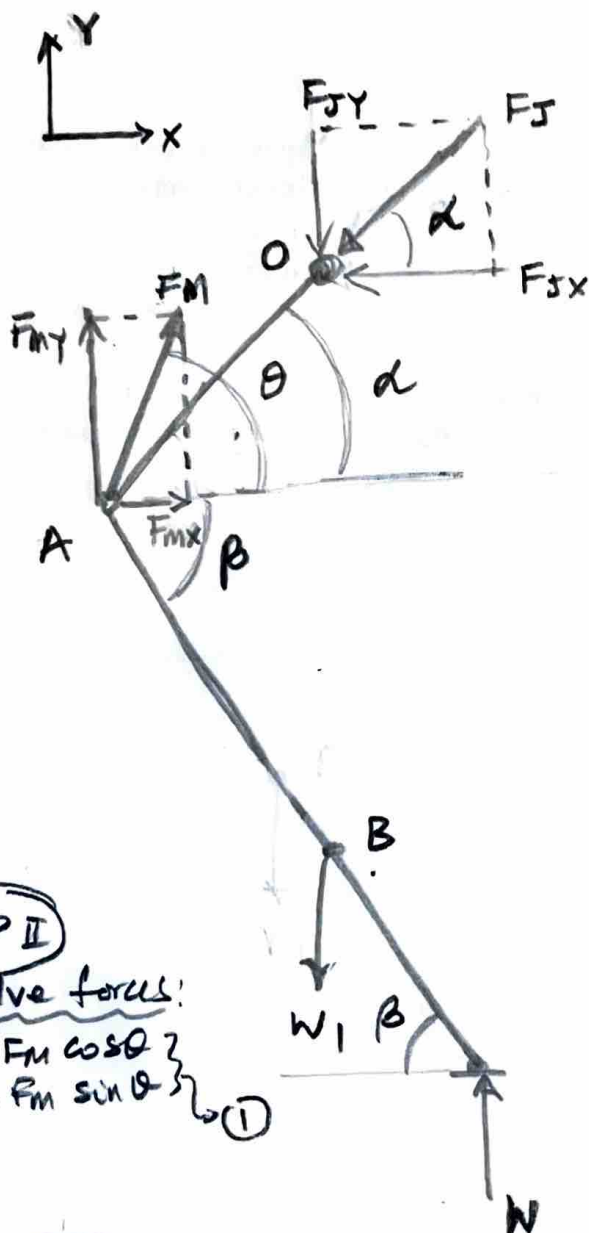
$$\rightarrow F_M = 2062.6 \text{ N}$$

$\rightarrow$  Line of action of  $F_M$  makes  $\theta = 69^\circ$  with horizontal

$\rightarrow$  Angle of femoral neck with horizontal:  $\alpha = 43^\circ$

$\rightarrow$  angle of femoral shaft:  $\beta = 79^\circ$

STEP I FBD:



STEP II

Resolve forces:

$$\begin{aligned} F_{mx} &= F_m \cos \theta \\ F_{my} &= F_m \sin \theta \end{aligned} \quad \text{--- (1)}$$

STEP II

MOMENT BALANCE  
about O:

$$\sum M_O = 0$$

STEP III

Resolve distances:

$$\begin{aligned} a_x &= a \cos \alpha \\ a_y &= a \sin \alpha \\ b_x &= b \cos \beta \\ c_x &= c \cos \beta \end{aligned} \quad \text{--- (2)}$$

{ Assume clockwise moments are positive }

$$a_y F_{my} - a_x F_{mx} - (c_x - a_x) W_1 + (b_x - a_x) W_1 = 0 \rightarrow (3)$$

From (1), (2) & (3) we have

$$(a \cos \alpha)(F_m \sin \theta) - (a \sin \alpha)(F_m \cos \theta) - (c \cos \beta - a \cos \alpha) W_1 + (b \cos \beta - a \cos \alpha) W_1 = 0$$



$$W = \frac{(a \cos \alpha)(F_m \sin \theta) - (a \sin \alpha)(F_m \cos \theta) + (b \cos \beta - a \cos \alpha) W_1}{(c \cos \beta - a \cos \alpha)}$$

$$\Rightarrow W = \frac{(a \cos \alpha)(F_m \sin \theta) - (a \sin \alpha)(F_m \cos \theta) + b \cos \beta W_1 - a \cos \alpha W_1}{(c \cos \beta - a \cos \alpha)}$$

$$W = \frac{(a \cos \alpha)[F_m \sin \theta - W_1] - (a \sin \alpha)(F_m \cos \theta) + b \cos \beta W_1}{(c \cos \beta - a \cos \alpha)}$$

↳ (4)

$$\begin{aligned} & (0.086 \cos 43^\circ)[2062.6 \sin 69^\circ - 125] \\ & - (0.086 \sin 43^\circ)(2062.6 \cos 69^\circ) \\ & + (0.343 \cos 79^\circ)(125) \end{aligned}$$

$$W = \frac{(0.894 \cos 79^\circ) - (0.086 \cos 43^\circ)}{0.1705 - 0.0628}$$

$$W = \frac{(0.0629)[1925.61 - 125] - 43.35 + 8.18}{0.1705 - 0.0628}$$

$$W = \frac{113.26 - 43.35 + 8.18}{0.107}$$

$$\Rightarrow \boxed{W \approx 729.8 \text{ N}}$$

Ans 1 ✓



STEP V

Apply equilibrium conditions

$$\boxed{\sum F_x = 0}$$

$$F_{Jx} = F_{Mx} = F_M \cos \theta \rightarrow (6)$$

$$\Rightarrow F_{Jx} = (2062.6) \cos 69^\circ$$

$$\Rightarrow F_{Jx} = 739.17 \text{ N}$$

$$\boxed{\sum F_y = 0}$$

$$F_{Jy} = F_{My} + N - N_1 \rightarrow (7)$$

$$= F_M \sin \theta + 729.8 - 125$$

$$= 2062.6 \sin 69^\circ + 729.8 - 125$$

$$= 2530.40 \text{ N}$$

$$\boxed{F_J = \sqrt{F_{Jx}^2 + F_{Jy}^2}} \rightarrow (8)$$

$$F_J = \sqrt{739.17^2 + 2530.40^2}$$

$$\Rightarrow \boxed{F_J = 2636.15 \text{ N}} \rightarrow (9)$$

← Ans 2

Angle made by reaction force ( $\phi$ )

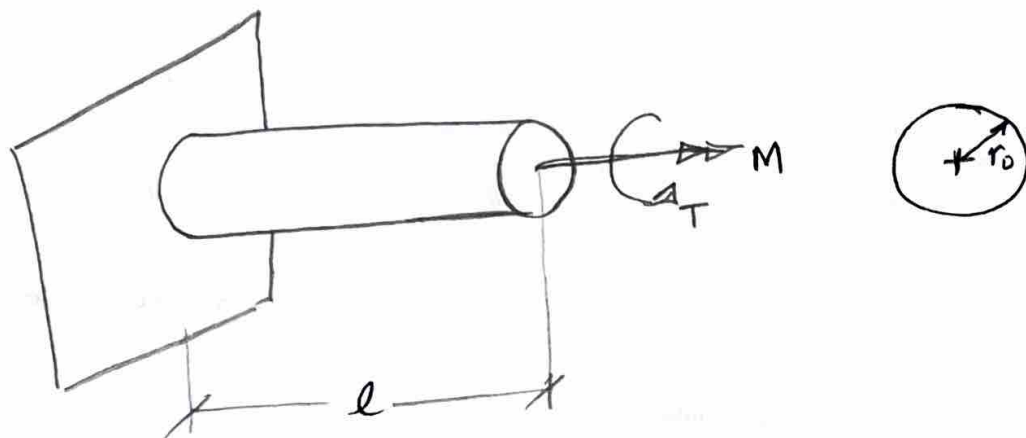
$$\boxed{\phi = \tan^{-1} \left( \frac{F_{Jy}}{F_{Jx}} \right)} = \tan^{-1} \left( \frac{2636.15}{739.17} \right)$$

$$\boxed{\phi = 74.33^\circ}$$



(9)

# **PROBLEM 4**



Given:  $l = 10 \text{ cm}$   
 $r_0 = 2 \text{ cm}$   
 $G = 10 \text{ GPa}$   
 $M = 3000 \text{ Nm}$

①  $J = \frac{\pi r_0^4}{2}$  for cylinder

$\Rightarrow J = \frac{\pi (2 \times 10^{-2})^4}{2} \text{ m}^4$

$J = 0.25 \times 10^{-6} \text{ m}^4$

Ans 1

Assume LEHI

Calculate:

- ①  $J \rightarrow$  polar moment of Inertia
- ② Maximum angle of twist ( $\theta$ )
- ③ Maximum shear stress ( $\tau$ ) in the transverse plane
- ④ Maximum shear strain in the transverse ( $\gamma$ ) plane

③ Max. shear stress:  $\tau_{\max} \Big|_{r=r_0} = \frac{M r_0}{J}$

$\Rightarrow \tau_{\max} = \frac{(3000 \text{ Nm}) (0.02 \text{ m})}{0.25 \times 10^{-6} \text{ m}^4}$

$\Rightarrow \tau_{\max} = 240 \text{ MPa}$

Ans 3

(2) Angle of twist:

$$\theta = \frac{Ml}{GJ}$$

$$\text{OR } \theta = \frac{l}{r_0} \gamma$$

shear strain

$$\begin{aligned} \theta &= \frac{(3000 \text{ Nm})(10 \times 10^{-2} \text{ m})}{(10 \times 10^9 \text{ Pa})(0.25 \times 10^{-6} \text{ m}^4)} \\ &= \frac{(3000 \text{ Nm})(0.1 \text{ m})}{(10 \times 10^9 \frac{\text{N}}{\text{m}^2})(0.25 \times 10^{-6} \text{ m}^4)} \end{aligned}$$

$$= \frac{(3000)(0.1)}{(10 \times 10^9)(0.25 \times 10^{-6})} = 3 \times 10^{-2} \times 4$$

$$\theta = 12 \times 10^{-2} \text{ radians} \Rightarrow \boxed{\theta = 0.12 \text{ rad}} \quad \text{Ans 2}$$

(4) Max. shear strain:

$$\gamma = \frac{\tau_{\max}}{G}$$

$$\begin{aligned} \Rightarrow \gamma &= \frac{240 \times 10^6 \text{ Pa}}{10 \times 10^9 \text{ Pa}} \\ &= 240 \times 10^{-4} \end{aligned}$$

$$\Rightarrow \boxed{\gamma = 0.024} \quad \text{Ans 4}$$

Alternate method  
for angle of twist ( $\theta$ )

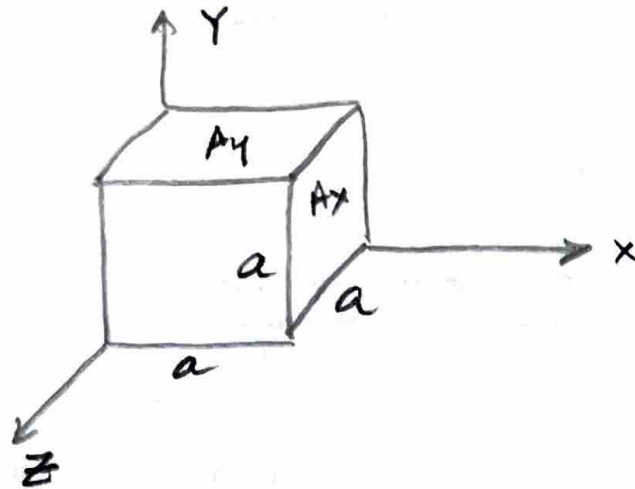
$$\theta = \frac{l}{r_0} \gamma$$

$$\theta = \frac{(10 \times 10^{-2} \text{ m})(240 \times 10^{-4})}{(0.25 \times 10^{-6} \text{ m})}$$

$$\theta = 120 \times 10^{-3} \text{ rad}$$

$$\Rightarrow \boxed{\theta = 0.12 \text{ rad}}$$

## Problem 5



Given:

$$a = 10 \text{ cm}$$

$$F_{xx} = F_{yy} = 2 \times 10^6 \text{ N}$$

$$E = 2 \times 10^{11} \text{ Pa}$$

$$\nu = 0.3$$

Assume LEHI model

Determine strains in x, y & z directions is:

- ①  $F_x$  &  $F_y$  are tensile
- ②  $F_x$  is tensile &  $F_y$  is compressive
- ③  $F_x$  &  $F_y$  are compressive

STEP I

$$A_x = A_y = a^2 = 100 \text{ cm}^2 = 1 \times 10^{-2} \text{ m}^2 \rightarrow \textcircled{1}$$

$$\tau_{xx} = \tau_{yy} = \frac{F_{xx}}{A_x} = \frac{F_{yy}}{A_y} = \frac{2 \times 10^6 \text{ N}}{1 \times 10^{-2} \text{ m}^2} = 2 \times 10^8 \text{ Pa} \rightarrow \textcircled{2}$$

From LEHI model:

$$\epsilon_{xx} = \frac{1}{E} [\tau_{xx} - \nu(\tau_{yy} + \tau_{zz})]$$

$$\epsilon_{yy} = \frac{1}{E} [\tau_{yy} - \nu(\tau_{zz} + \tau_{xx})]$$

$$\epsilon_{zz} = \frac{1}{E} [\tau_{zz} - \nu(\tau_{xx} + \tau_{yy})]$$

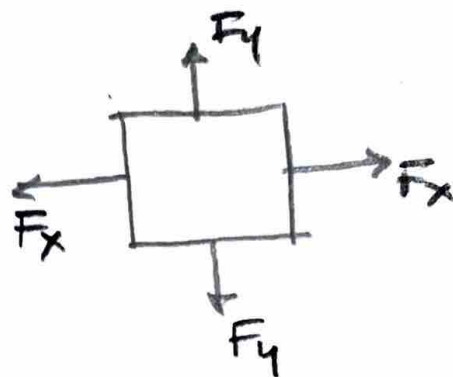
③



①  $F_x, F_y$  are tensile :

$$\tau_{xx} = \tau_{yy} = \tau \quad \left\{ \begin{array}{l} \text{We know} \\ \text{this} \end{array} \right.$$

$$\tau_{zz} = 0$$



From Q(5):

$$\epsilon_{xx} = \frac{1}{E} [\tau_{xx} - \nu(\tau_{yy} + \tau_{zz})]$$

$$\Rightarrow \epsilon_{xx} = \frac{1}{E} [\tau_{xx} - \nu \tau_{yy}]$$

$$\Rightarrow \boxed{\epsilon_{xx} = \left(\frac{1-\nu}{E}\right) \tau} \Rightarrow \epsilon_{xx} = \left(\frac{1-0.3}{2 \times 10^{11} \text{ Pa}}\right) (2 \times 10^8 \text{ Pa})$$

$$\epsilon_{yy} = \frac{1}{E} [\tau_{yy} - \nu(\tau_{xx} + \tau_{zz})] \Rightarrow \boxed{\epsilon_{xx} = 0.7 \times 10^{-3}}$$

$$\Rightarrow \epsilon_{yy} = \frac{1}{E} [\tau_{yy} - \nu \tau_{xx}]$$

$$\Rightarrow \boxed{\epsilon_{yy} = \left(\frac{1-\nu}{E}\right) \tau} \Rightarrow \boxed{\epsilon_{yy} = 0.7 \times 10^{-3}}$$

$$\epsilon_{zz} = \frac{1}{E} [\tau_{zz} - \nu(\tau_{xx} + \tau_{yy})]$$

$$\Rightarrow \epsilon_{zz} = \frac{1}{E} [-\nu(2\tau)]$$

$$\Rightarrow \boxed{\epsilon_{zz} = -\frac{2\nu}{E} \tau} \Rightarrow \epsilon_{zz} = -\frac{2(0.3)(2 \times 10^8 \text{ Pa})}{2 \times 10^{11} \text{ Pa}}$$

$$\Rightarrow \boxed{\epsilon_{zz} = -0.6 \times 10^{-3}}$$

(2)  $F_x$  is tensile,  $F_y$  is compressive

$$\tau_{yy} = -\tau_{xx} \quad \left. \begin{array}{l} \text{We know} \\ \text{this} \end{array} \right\}$$

$$\tau_{zz} = 0$$

$$(\tau_{xx}, -\tau_{yy})$$

$$\epsilon_{xx} = \frac{1}{E} [\tau_{xx} - \nu (\tau_{yy} + \tau_{zz})]$$

$$\Rightarrow \epsilon_{xx} = \frac{1}{E} [\tau_{xx} - \nu (-\tau_{xx})]$$

$$\Rightarrow \epsilon_{xx} = \left( \frac{1+\nu}{E} \right) \tau \quad \left\{ \text{where } \tau_{xx} = \tau \right\}$$

$$\Rightarrow \epsilon_{xx} = \left( \frac{1+0.3}{2 \times 10^{11} \text{ Pa}} \right) (8 \times 10^8 \text{ Pa}) \Rightarrow \epsilon_{xx} = 1.3 \times 10^{-3}$$

$$\epsilon_{yy} = \frac{1}{E} [\tau_{yy} - \nu (\tau_{xx} + \tau_{zz})]$$

$$= \frac{1}{E} (-\tau_{xx} - \nu \tau_{xx})$$

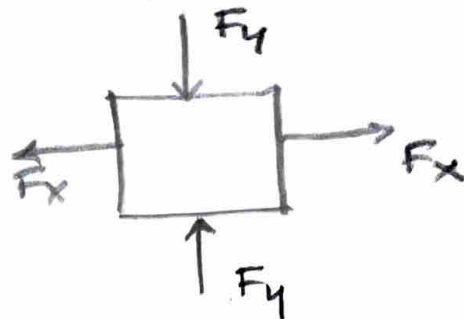
$$\epsilon_{yy} = - \left( \frac{1+\nu}{E} \right) \tau \quad \left\{ \text{where } \tau_{xx} = \tau \right\}$$

$$\Rightarrow \epsilon_{yy} = -1.3 \times 10^{-3}$$

$$\epsilon_{zz} = \frac{1}{E} [\tau_{zz} - \nu (\tau_{xx} + \tau_{yy})]$$

$$\Rightarrow \epsilon_{zz} = \frac{1}{E} [-\nu (\tau_{xx} - \tau_{xx})]$$

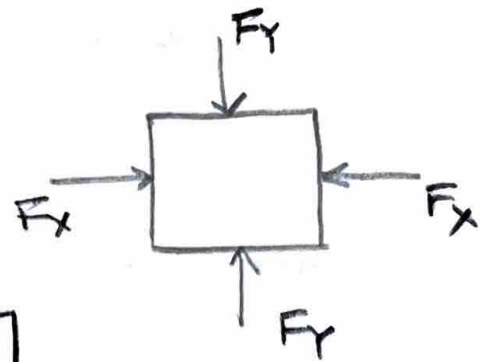
$$\Rightarrow \epsilon_{zz} = 0$$



③  $F_x, F_y$  are compressive :

$$(-\tau_{xx}, -\tau_{yy})$$

$$\left. \begin{aligned} -\tau_{xx} &= -\tau_{yy} \\ \tau_{zz} &= 0 \end{aligned} \right\} \begin{array}{l} \text{We know} \\ \text{this} \end{array}$$



$$\epsilon_{xx} = \frac{1}{E} [\tau_{xx} - \nu(\tau_{yy} + \tau_{zz})]$$

$$\Rightarrow \epsilon_{xx} = \frac{1}{E} [-\tau_{xx} - \nu(-\tau_{yy})]$$

$$= \frac{1}{E} [-\tau_{xx} + \nu\tau_{xx}]$$

$$\boxed{\epsilon_{xx} = \left(\frac{\nu-1}{E}\right)\tau} \quad \left\{ \text{where } -\tau_{xx} = -\tau \right\}$$

$$\Rightarrow \epsilon_{xx} = \left(\frac{0.3-1}{2 \times 10^8 \text{ Pa}}\right) (2 \times 10^8 \text{ Pa}) \Rightarrow \boxed{\epsilon_{xx} = -0.7 \times 10^{-3}}$$

$$\epsilon_{yy} = \frac{1}{E} [\tau_{yy} - \nu(\tau_{xx} + \tau_{zz})]$$

$$= \frac{1}{E} [-\tau_{xx} - \nu(-\tau_{xx})] \Rightarrow \left(\frac{\nu-1}{E}\right)\tau = \epsilon_{yy}$$

$$\Rightarrow \boxed{\epsilon_{yy} = -0.7 \times 10^{-3}}$$

$\left\{ \text{where } \tau_{xx} = \tau \right\}$

$$\epsilon_{zz} = \frac{1}{E} [\tau_{zz} - \nu(\tau_{xx} + \tau_{yy})]$$

$$\Rightarrow \epsilon_{zz} = \frac{1}{E} [-\nu(-\tau_{xx} - \tau_{xx})]$$

$$\Rightarrow \boxed{\epsilon_{zz} = \frac{2\nu}{E}\tau} \quad \left\{ \text{where } \tau_{xx} = \tau \right\}$$

$$\Rightarrow \epsilon_{zz} = \left[\frac{2(0.3)}{2 \times 10^8 \text{ Pa}}\right] (2 \times 10^8 \text{ Pa}) \Rightarrow \boxed{\epsilon_{zz} = 0.6 \times 10^{-3}}$$